Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s 3 of the UGC Act, 1956)
Off Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043
Phone No. 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 12

Team No.: 25

Project Title: ALARAM CLOCK

Team Members

Roll Number	Student Name
2520080015	V.HARI PRIYA
2520080068	M.SAI HARIKA

1. Abstract

The Alarm Clock is an Operating Systems project designed to provide a simple and reliable way to set and manage alarms on a Linux/Ubuntu system. The application allows users to set a specific alarm time and continuously monitors the system clock. When the set time is reached, the system generates an alarm notification or sound to alert the user. The project demonstrates important Operating System concepts such as processes, system calls, signals, process synchronization, and time management. Linux APIs such as `fork()`, `sleep()`, and signal handling can be used to implement the alarm functionality. The project provides practical experience in process creation, scheduling, and communication between the application and the operating system. It offers a lightweight and user-friendly solution for managing reminders and alarms.

2. Problem Statement

People often forget important tasks, meetings, classes, or daily activities because they do not receive timely reminders. The objective of this project is to develop a simple Alarm Clock application on Ubuntu/Linux that allows users to set a specific alarm time and receive an alert when the scheduled time is reached. The system should continuously monitor the clock, manage the alarm efficiently, and notify the user at the correct time using Operating System concepts such as processes, signals, system calls, and process synchronization.

3. Objectives

1. To develop a simple and reliable alarm clock application on Ubuntu/Linux.

2. To allow users to set and manage alarm times easily.

3. To continuously monitor the system time and trigger the alarm at the scheduled time.

4. To use Operating System concepts such as processes, signals, system calls, and synchronization.

4. Proposed Methodology

The proposed Alarm Clock system will be developed on Ubuntu/Linux using Operating System concepts such as process creation, system calls, signals, and process synchronization. The user first enters the desired alarm time, and the system continuously monitors the current system time. A separate process can be created using fork() to manage the alarm while the main process continues its operation. The alarm process uses sleep() to wait until the specified time and checks the system clock. When the current time matches the alarm time, a signal is generated and the system produces an alert sound or notification. After triggering the alarm, the process terminates safely. This methodology provides a simple, efficient, and practical demonstration of Linux process and time-management concepts.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Creates and manages processes for alarm monitoring.
fork()	Creates a child process to handle the alarm.
sleep()	Makes the process wait until the required time.
Signals	Notifies the process when the alarm needs to be triggered.
signal() / sigaction()	Handles alarm-related signals
time()	Gets the current system time.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520080015	V.HARI PRIYA	Designed the alarm logic, implemented time monitoring, process creation using fork(), and signal handling
2520080068	M.SAI HARIKA	Developed the user interface, implemented alarm notifications/sound, tested the application, and prepared documentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system and development environment
C / C++	Used to develop the Alarm Clock application
GCC Compiler	Compiles and executes the C program
Linux System Calls / APIs	Used for process, time, and signal management
VS Code / Text Editor	Used for writing and editing source code

8. Expected Outcome

The project is expected to produce a simple and reliable Alarm Clock application on Ubuntu/Linux that allows users to set a specific alarm time and receive an alert when the scheduled time is reached. The system will demonstrate important Operating System concepts such as process creation, system calls, signals, time management, and process synchronization. The project will also provide practical experience in Linux programming and the use of C programming with Linux APIs.

9. GitHub Repository

GitHub Repository Link: https://github.com/veridhiharipriya/KLH-AI-DS-2026-25-AlarmClock

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: M. Raghupathi

Signature:

Remarks:

Faculty Use

Project Approved:

☐ Yes ☐ No

Date:

Faculty Signature: